

Liam Dubai

Ph.D. Candidate in Astronomy
The Ohio State University

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Education

The Ohio State University	Columbus, OH
Ph.D. Candidate in Astronomy	2024 – Present
M.S. in Astronomy	August 2024
Thesis Advisor: Prof. Jennifer A. Johnson	
Whitman College	Walla Walla, WA
B.A. in Physics-Astronomy, Music (Performance), <i>summa cum laude</i>	May 2021
Honors in Physics-Astronomy, Thesis Advisor: Prof. Andrea K. Dobson	
Honors in Music (Performance), Thesis Advisor: Dr. Sally Singer Tuttle	

Honors & Awards

Anne S. Tuttle Citizenship, Engagement, and Outreach Prize	2026
Recognizing exceptional effort and impact in outreach, public engagement, or service.	
OSU Presidential Fellowship	2025
Support for final-year students in recognition of outstanding scholarly accomplishments.	
NSF Graduate Research Fellowship Proposal Honorable Mention	2022
The Ohio State University Graduate Fellowship	2021 – 2022
Financial support for first-year graduate students.	
Whitman College Valedictorian Scholarship	2017

Research

Publications: 7 papers (4 first-author, 3 contributing) | [NASA ADS Library of Publications](#)
(A full publication list is included.)

Interests: Galactic chemical evolution, the Milky Way, origin of the elements, Type Ia supernovae, large stellar spectroscopic surveys

Seminars & Conference Presentations:

Invited Seminar	University of Notre Dame (South Bend, IN)	2026
Contributed Talk	Sloan Digital Sky Survey (SDSS-V) Collab. Meeting (Leuven, Belgium)	2026
Invited Seminar	Carnegie Observatories (Pasadena, CA)	2026
Poster	SDSS-V Collaboration Meeting (Heidelberg, Germany)	2025

Contributed Talk	SDSS-V Collaboration Meeting (Las Cruces, NM)	2024
Contributed Talk	Hayes Advanced Research Forum , Ohio State University	2024
Poster	Surveying the Milky Way: The Universe in Our Own Backyard , California Institute of Technology (Pasadena, CA)	2023
Contributed Talk	241st American Astronomical Society (AAS) Meeting (Seattle, WA)	2023
Poster	237th AAS Meeting (Virtual)	2021
Poster	235th AAS Meeting (Honolulu, HI)	2020

Teaching & Mentorship

Polaris Mentorship Course, The Ohio State University

<i>Instructor of record</i>	2025 – 2026
<i>Near-peer mentor</i> to 4 students	2022 – 2025
Undergraduate students work with graduate mentors to develop science identity, build community, and complete a guided research project and poster presentation.	
Website: https://u.osu.edu/polaris	

Teaching Coursework, The Ohio State University

Astronomy 7193: College Teaching in Astronomy	Summer 2025
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Python Coding Workshops Lead Instructor, The Ohio State University

2024 – 2025
Organized lectures and developed engaging challenge problems for undergraduates.

Graduate Teaching Assistant, The Ohio State University

Astronomy 3350: Methods of Astronomical Observation & Data Analysis	2022 – 2023
Astronomy 1221: Astronomy Data Analysis (2 semesters)	Fall 2023
Astronomy 1101: Planets to Cosmos Lab (2 sections)	Spring – Fall 2023
	Fall 2022

Astronomy Teaching Assistant & Tutor, Whitman College

Astronomy 177: Sky and Planets	2019 – 2020
Astronomy 179: Galaxies and Cosmology	Spring 2020
Astronomy 178: Sun and Stars	Fall 2019
	Spring 2019

Service & Outreach

Polaris Leadership Committee, The Ohio State University

2023 – Present
Graduate student-led organization fostering community and improving retention of non-traditional and underrepresented students in physics and astronomy.

“Galaxy Hour” Meeting Co-Organizer, Ohio State Astronomy Dept.

2023 – 2026
Organized weekly topical research meeting and visiting speaker seminars.

Planetarium Presenter, Arne Slettebak Planetarium

2021 – 2025
The Ohio State University, Columbus, OH
Lead presenter for over 100 planetarium shows.
Audiences include: K-12, boy & girl scouts, university groups, and the general public.

Publications

Dubay, L. O. et al. (16 additional authors) 2026, “A Broken Clock Is Right Twice a Day: [Ce/Mg] Is Not a Universal Chemical Clock”, *arXiv e-prints*, [arXiv:2608.26149](#)

SDSS Collaboration et al. (inc. **Dubay, L.**) 2026, “The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper”, *arXiv e-prints*, [arXiv:2607.26149](#)

Roberts, J. D., Pinsonneault, M. H., Johnson, J. A., **Dubay, L. O.**, & Johnson, J. W. 2026, “[C/N] Ages for Red Giants and their Implications for Galactic Archaeology”, *The Astrophysical Journal* 1002, 191, doi:[10.3847/1538-4357/ae5bae](#)

Dubay, L. O., Johnson, J. A., Johnson, J. W., Roberts, J. D. 2026, “Challenges to the Two-Infall Scenario by Large Stellar Age Catalogs”, *The Astrophysical Journal* 1000, 244, doi:[10.3847/1538-4357/ae4c53](#)

Johnson, J. W. et al. (inc. **Dubay, L. O.**) 2025, “The Milky Way Radial Metallicity Gradient as an Equilibrium Phenomenon: Why Old Stars Are Metal Rich”, *The Astrophysical Journal* 988, 8, doi:[10.3847/1538-4357/addbe5](#)

Dubay, L. O., Johnson, J. A., & Johnson, J. W. 2024, “Galactic Chemical Evolution Models Favor an Extended Type Ia Supernova Delay-Time Distribution”, *The Astrophysical Journal* 973, 55, doi:[10.3847/1538-4357/ad61df](#)

Dubay, L. O., Tucker, M. A., Do, A., Shappee, B. J., & Anand, G. S. 2022, “Late-Onset Circumstellar Medium Interactions are Rare: An Unbiased *GALEX* View of Type Ia Supernovae”, *The Astrophysical Journal* 926, 98, doi:[10.3847/1538-4357/ac3bb4](#)